Muscle Model · Staff Answers

Learning goal: I can test a card and string arm model to show how two muscles pull a joint.

Arrival Answers

1. They stand for bones.
2. It stands for the elbow joint.
3. They stand for two muscles.
4. Yes. You can ask or pass.

Starter & Hinge Answers

Starter: A. Just like a muscle, a string can pull a bone, but it cannot push.

Hinge: A. If both pull at once, they pull against each other and the joint stays stuck.

Evidence Desk Cards

Stiff card strips: Bone: stiff card strips stand for the upper and lower arm bones.

Split pin in the corner: Joint: the pin acts like the elbow hinge joint.

Top String A: Top muscle: pulling String A bends the arm up.

Back String B: Back muscle: pulling String B makes the arm straight.

Push on String A: Model result: the string goes slack and the card does not move.

Hand pulling the string: Model limit: real muscles get

short by themselves inside the arm.

Check & Repair Answer

Strings and muscles cannot push. Pull back String B to straighten the arm.

Exit Ticket Answers

1. Muscle. 2. The lower card arm bent up. 3. A real muscle gets short and fat on its own without a hand pulling its end.